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Root Exudation Facilitates Water Infiltration and Rewetting of Dry Soil

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ABSTRACT

The way plant roots facilitate water infiltration in soil may be just as important as the efficiency with which the root system in turn extracts it from the soil. Here we studied the mechanisms through which the root system facilitates water infiltration through a dry soil layer. Dye tracing experiments were conducted in model soil microcosms to characterise how root growth and exudation affect the permeability of dry layers of the model soil. Results showed that the growth of plant roots through the dry layers of an artificial soil increased the water infiltration rate. In the absence of roots, dissolved root exudates had a significant effect on water infiltration but penetration of the dry layer by a needle did not. We conclude that in dry soil, root architecture and root exudation act synergistically to increase soil hydraulic conductivity, and this may help decrease the water lost by evaporation. This mechanism could be used to select root traits that match the soil and climate, thereby improving water use efficiency in agriculture.

1 | Introduction

Hydrologists have long acknowledged the significant role of vegetation in influencing water movement in soil, and as Clothier and Green aptly expressed, the roots are the "big movers" (Clothier and Green 1997). A primary driver is the extraction of water from the soil through root uptake and subsequent transpiration, which not only reduces soil water content but also causes water to move along gradients of the water potential. However, before uptake, the roots also influence water transport through modifications of the soil hydraulic properties. The presence of plant roots in soil has been linked to preferential flow of water in dye-tracing experiments. For example, Jačka et al. (2021) observed increased dye concentration around plant roots in a tracer experiment. This finding was corroborated by measurements of saturated hydraulic conductivity in soil column experiments, which showed that the soil conductivity increased due to the presence of roots (Leung 2018). Various studies have also demonstrated that the hydraulic conductivity increases with the root length density (Xiao et al. 2024). In field experiments, it was also observed that most types of vegetation increase the infiltration rate of bare soil (Marshall et al. 2014; Song et al. 2017), although in some cases, roots also growth through biopores (Zhou et al. 2021) and this may link to a temporary reduction in the hydraulic conductivity of the soil.

Changes in soil hydraulic properties can be attributed to physical modifications in the soil's pore structure. As roots grow, they displace the surrounding soil, creating a cavity and rearranging the distribution of the surrounding soil pores (Dexter 1987). Because the size and connectivity of soil pores determine the resistance of the flow of water, these changes in soil porosity in turn affect the permeability of the soil (Schulz et al. 2019). In the study of (Anselmucci et al. 2021) roots grown in sand created an increase in porosity because of dilation. In more structured soil however, the expansion of the root cavity compacted the surrounding soil, and this can lead to a reduction in the soil porosity (Bruand et al. 1996; Helliwell et al. 2017; Koebernick et al. 2019), although this usually occurs in the bulk soils in which initial porosity is low (Lucas et al. 2019). Other factors such as the shrinking or decay of the root can also create macropores (Ni et al. 2019; Duddek et al. 2022) where the flow of water can occur with least resistance.

Chemical modifications of the soil solution caused by rhizodeposition influence the properties of the soil solution as well, although the properties affected can have an antagonistic effect. High molecular weight polymers such as mucilage increase the viscosity of the soil solution (Read et al. 1999) which in turns reduces the flow speed and increases the resistance to water movements. Mucilages can be gelatinous because of polymers that are hydrophilic, crosslinked, and can absorb and retain a significant amount of water (Naveed et al. 2019). This property is critical to maintain moisture around the root when the soil is drying (Carminati et al. 2010). Once dry however, root rhizodeposits can become water repellent and may slow down water infiltration during rewetting (Ahmed et al. 2016). Rhizodeposits also contain surfactants, and this likely facilitates water movement during infiltration and the rewetting of dry and water repellent soils (Read et al. 1999; Read et al. 2003). Root exudation was also found to be strongly affected by water stress (Williams and de Vries 2020).

To understand and potentially harness root-driven soil water redistribution, it is essential to understand how root growth modifies the hydraulic properties of the rhizosphere. However, progress towards this goal has been hindered by the number of properties affected by root activity. This study aims to determine whether physical or chemical modifications of the soil substrate exert a greater influence on soil water infiltration. We carried out dye tracing experiments in model transparent soil microcosms containing dry hydrophobic layers. Unlike classic dye tracing experiment, the use of transparent soil as model substrate allowed precise quantification of the quantity of dye percolating through depth, and measurement of the rewetting of the dry layer. Mathematical models were then used to quantify the respective role of root exudation and root growth on the permeability of these hydrophobic layers.

2 | Materials and Methods

2.1 | Plant Material and Germination

Seeds of winter wheat (*Triticum aestum var. Filon*) were first hydrated for 1 h in sterilised water and then sterilised for 15 min in a 1% calcium hypoclorite solution (042548.30, Thermo Scientific, Spain). After numerous washes, seeds were placed in distilled water for two more hours before transfer to agar plates in an incubator at 23°C. The incubation time was 3 days when running microcosm experiments and 5 days for root exudate extraction experiments.

2.2 | Extraction of Root Exudates

Obtaining an extract of root exudates that represent adequately what is being released in soil is challenging (Oburger and Jones 2018). The method used is based on plants grown in hydroponic solutions where extraction is done in distilled water. Although the method is less suited to the quantitative estimation of exudation rates, it has successfully been employed for the physical characterisation of root exudates (Naveed et al. 2019). This has enabled the distinction between gelatinous and surfactant effects and facilitated the characterisation of differences in the physical properties of root exudates among various plant species. Seedlings whose radicles had reached 3–5 cm were placed on a 3D printed punched plate (5 mm diameter holes arranged on a regular grid with 2 cm spacing). Each plate hosted approximately 50 seeds and was placed on top of a $15 \times 21 \times 7$ glass dish (Ikea, Spain) with roots dipping into 1 L of Hoagland nutrient solution (092621822, MP Biomedicals, Spain) adjusted to a pH of 7 using potassium hydroxide (KOH). Plants grew in a growth chamber (23°C, 60% humidity with 14/10 h day/night cycle) with the nutrient solution aerated using a pneumatic pump (Souslow, Spain) during 7 days. The glass dishes used for the hydroponic culture were transparent. To suppress contaminant growth such as algae, and prevent light-induced root responses, the outside of the glass dish was painted in black. Currently, all methods for extraction of soluble exudates possess some limitation After 6 days of growth, the nutrient solution was replaced with 0.5 L of distilled water. Roots grew 24 h in distilled water before three 50 mL falcon tubes (EP0030122232, Sigma-Aldrich, Spain) were used to collect 120 mL of solution from each glass dish. The root exudate solutions were then frozen at −80°C. To adjust the concentration of root exudates, falcon tubes containing root exudate solutions were freeze-dried and 2 mL of distilled water were subsequently added before pooling all the root exudates from one glass dish into a 15 mL falcon tube (EP0030122194, Sigma-Aldrich, Spain). The root exudate solution obtained was freeze-dried a second time for weighing, and distilled water was added to adjust the concentration of the root exudate solution to 1 mg mL^{-1}. Root exudates were then stored frozen at −20°C. We ran 4 experiments with a total of 8 plates, Root exudates were pooled to ensure uniform chemical properties across all water infiltration experiments.

2.3 | Characterisation of Surface Tension Using the Pendent Drop Method

Root exudates solutions of 1 mg mL^{-1} and 10 mg mL^{-1} concentrations were characterised together, and distilled water was used as control. Drops of solutions with a volume of approximately 170 µL were obtained using a syringe pump (Pump 11 Pico Plus Elite, Harvard Apparatus, United States) and a blunt end dispense tips with an inner diameter of 0.819 mm (Fisnar QuantX, Ellsworth Adhesive, Spain). Images were acquired with a CMOS camera (DCC1545M, Thorlabs, UK) and a telecentric 1× objective lens (375-036-2, Mitutoyo, Germany). Images of pendent drops were acquired and subsequently analysed using the OpenDrop software (Berry et al. 2015). Each measurement was repeated 12 times.

2.4 | Transparent Soil

Transparent soil was prepared as described earlier (Downie et al. 2012). Briefly, Nafion pellets (NR50 1100, Ion Power Inc, USA) were fractured to a texture similar to sand (0.25–1.25 mm) using a freezer mill (6850 Freezer/Mill, SPEX CertiPrep, UK) and a series of sieves. The pH of the substrate was then adjusted by washing with Hoagland nutrient solution. With a density for Nafion of approximately 1.9 g cm^{-3} and a porosity of 0.3, the estimated substrate density is 1.58 g cm^{-3}.

2.5 | Microcosm Preparation

Microcosm chambers were assembled following the protocol described by Liu et al. (2025) from microscope glass slides ($76 \times 26 \times 1$ mm^3, VWR, Spain). Glass slides were bonded by 4 mm polydimethylsiloxane spacer (PDMS, SYLGARD 184, Sigma-Aldrich, Spain) following surface treatment by oxygen plasma for 15 s at 100 W (HPT-100, Henniker Plasma, UK). After surface treatment, the glass and PDMS surfaces were brought together, and pressure was applied to ensure a strong bond between all surfaces in contact. Following the assembly of the different parts, the space available for placing plants and soil is $68 \times 18 \times 4$ mm^3. The microcosm was then filled with transparent soil layers with different water content (Figure 1A). The bottom 1.5 cm layer (approximate dry weight of 1.7 g) of the microcosm chamber was filled with transparent soil fully saturated with water. Then a 0.8 cm layer of dry transparent soil was added (approximate dry weight of 0.9 g). The dry transparent soil was obtained by heating in an oven at 105°C for 24 h. Since Nafion is very hydrophobic when dry (Goswami et al. 2008) the rewetting of the dry layer by water only is extremely slow. The top layer was 1 cm thick (approximate dry weight of 1.1 g) of transparent soil fully saturated with water.

2.6 | Dye Tracing Experiment

First, we studied how the presence of plant roots affected the permeability of the dry transparent soil layer (experiment 1, Supporting Information S1: Figure S1). Control samples

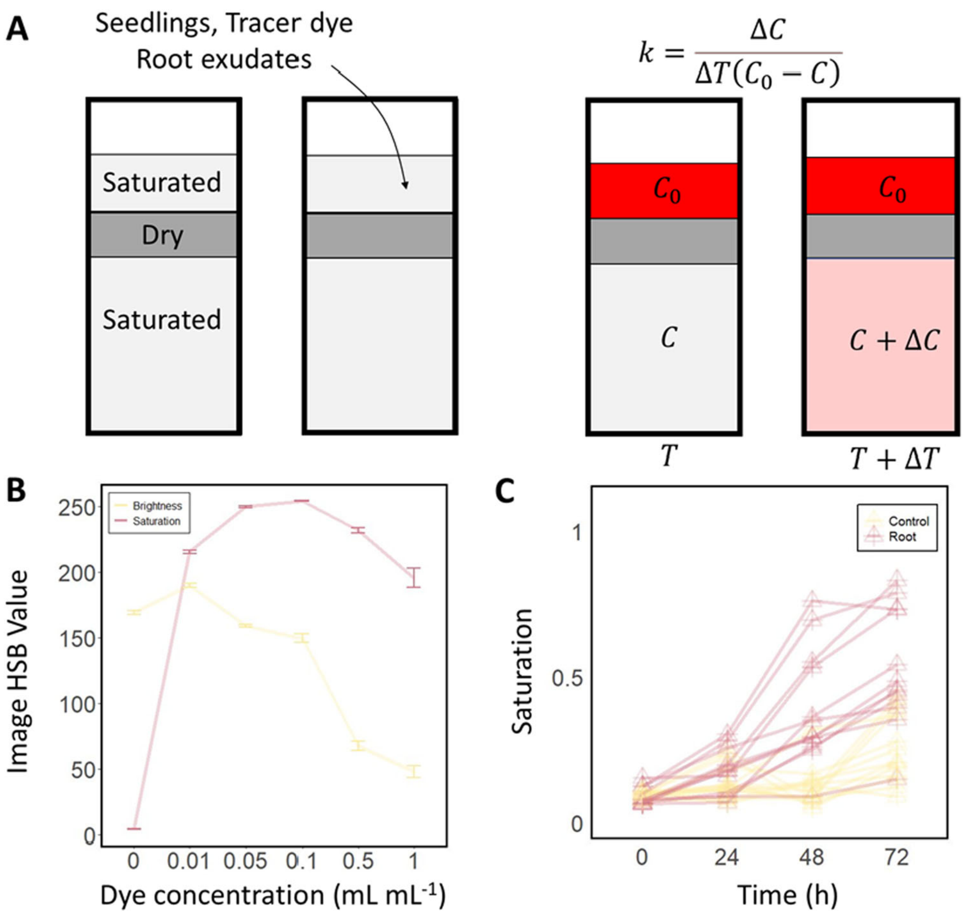

FIGURE 1 | Model system for quantifying the impact of root exudates on water transport (A) Diagram of the microcosm system. At the start of the experiment, the microcosm is comprised of three layers of transparent soil (left). The top layer of transparent soil is saturated with water to allow fast diffusion of the dye. The middle layer is oven dried to produce a hydrophobic barrier. The bottom layer is saturated with water for precise determination of the quantity of dye moving through the dry layer. During the experiment (right) the dye was introduced at T0 at the top of the microcosm chamber and diffused rapidly through the top layer of transparent soil. At time T1 the tracer dye became visible, and a permeability coefficient could be calculated. (B) The dye concentration in the liquid can be quantified from the image saturation value (red) at low dye concentration, and with the image brightness value (yellow) at high concentration. (C) Example of time lapse data of the saturation value (0–1) captured in the bottom transparent soil layer.

consisted of microcosm chambers prepared as described in the previous section but without seedlings transferred into them. Wheat seedlings were transferred when the root length was 0.5–0.8 mm in length. All microcosm chambers were closed with parafilm tape (HS234526B, Sigma-Aldrich, Spain) and placed in a growth chamber at 23°C, 60% humidity with 14/10 h day/night cycle. Distilled water was added to the samples using a syringe to maintain the top layer of transparent soil saturated. The tracer dye was added 8 days after the transfer of the root seedlings. A food dye (red food colourant, Vahiné, Spain) was diluted to 0.1 mL mL^{-1} to obtain the tracer dye solution. 0.2 mL of that solution was then added on the top of the microcosm chamber. Half an hour later, the first image was taken ($t = 0$ h). The following images were taken 24, 48 and 72 h after the first image. The experiment was repeated 3 times with a total of 15 samples with roots and 13 controls samples without roots. Two root samples were discarded because of leakage of the microcosm chamber.

In the second experiment (experiment 2, Supporting Information S1: Figure S1), we quantified the effect of root exudates on the infiltration of water. In this case the tracer dye solution was diluted with the root exudate solution so that resulting root exudate concentration was approximately 0.9 mg mL. Then 0.2 mL of that solution was then added at the top of the microcosm (concentration of root exudate approximately 0.16 mg g^{-1}). All microcosm chambers were sealed with parafilm tape and placed in the growth chamber under the previously described conditions. The first images were obtained half an hour after inoculation ($t = 0$ h). The following images were obtained 72, 96 and 120 h after the first image. The experiment was repeated three times with a total of 19 samples containing root exudates and 16 control samples containing only water. One control sample was discarded due to a leak.

In a third experiment (experiment 3, Supporting Information S1: Figure S1), we studied the effect of the presence of root exudates in the dry transparent soil layer. To prepare dry transparent soil containing root exudates, 5 g of wet transparent soil was mixed with 1 mL of the root exudate solution. The wet transparent soil was then left to dry at 105°C for 48 h in an oven. Therefore, the resulting root exudate concentration in the dry transparent soil was similar to that of the top layer, 0.633 mg g^{-1}. The experiment involved two independent variables: the presence of exudates in the top layer and the presence of exudates in the middle layer. Therefore, four treatments were considered, namely, samples that did not contain root exudates (NN), samples that contained root exudates in the middle layer (NE), samples that contained root exudates in the top layer EN), and samples that contained root exudates in both layers (EE). All microcosm chambers were sealed with parafilm tape and placed in the growth chamber. The first image was acquired half an hour after introduction of the tracer dye ($t = 0$ h). The following images were taken 72, 96 and 120 h after the first image. The experiment was repeated three times with a total of eight samples for each combination.

Finally, in the fourth experiment (experiment 4, Supporting Information S1: Figure S1), we studied the effect of the rearrangement of transparent soil particles induced by penetration of the dry middle layer. A hypodermic needle (12684536, Fisher Scientific, Spain) was pushed approximately 2.5 cm into the transparent soil. The needle was 0.8 mm in diameter which is similar to the diameter of wheat roots (Hallam et al. 2021). Control samples were not penetrated by the hypodermic needle. 0.2 mL of the tracer dye solution was added at the top of the microcosm chamber. All microcosm chambers were sealed with parafilm tape and placed in the growth chamber. Half an hour after inoculation of the tracer dye, the first image was taken ($t = 0$ h). The following images were taken 24, 48 and 120 h after the first image. The experiment was repeated three times with a total of 15 replicates for the treatments and 15 replicates for control treatment. One sample was discarded due to a leak.

2.7 | Quantitative Image Analysis

Images of microcosm chambers were acquired using a Canon EOS 2000D camera (Canon, Spain) with an exposure time of 1/1000 s, ISO-400 speed, focal length of 39 mm and a f-stop of f/5.6. Colour images were stored uncompressed in colour format (sRGB) at resolution of 6000×4000 pixels. To quantify the tracer dye concentration from image data, calibration samples with dye concentration of 0, 0.01, 0.05, 0.1, 0.5 and 1 mL mL^{-1} were prepared in triplicate. Images were acquired for each sample and relationships were established between the dye concentration and the image brightness, and between dye concentration and the image saturation value (Figure 1B). To quantify dye concentration in the microcosm chambers containing transparent soil layers, regions of the image corresponding to the top, middle and bottom layers of transparent soil were cropped from the raw image data. Images were transformed to hue, saturation and brightness (HSB) format and the saturation value was used to determine the dye concentration (Figure 1C). All image processing tasks were performed using the ImageJ software (Schmid et al. 2010). The image data set generated is available on Figshare under the following DOIs https://doi.org/10.6084/m9.figshare.29194898.v1 and https://doi.org/10.6084/m9.figshare.29136761.v1.

The permeability of the dry transparent soil layer k [s^{-1}] was calculated using the following formula

$$k = \frac{\Delta C}{\Delta T(C_0 - C)} \tag{1}$$

Here, C_0 and C are the dye concentrations at time T in the top and bottom layer respectively, and $C + \Delta C$ is the dye concentration in the bottom layer at time $T + \Delta T$.

2.8 | Statistical Analysis

Statistical models were constructed to analyse the effects of plant roots, root exudates or to analyse the effect of the needle on both the permeability k and the tracer dye concentration in the dry transparent soil layer. We used generalised linear models to correlate the permeability of the dry transparent soil layer with both time and treatment effects. We tried different variance functions and found that heterogeneity was best corrected using variance covarying with treatments rather than with time. Model selection was based on the Akaike information criterion (AIC). Statistical analyses were performed with R (R Core Team 2021) using the glm package (Dobson and Barnett 2018).

2.9 | Model of Water Infiltration and Rewetting

To quantify the role of root exudates in infiltration and rewetting an equation for the transport of water in the transparent soil is coupled with equations for the diffusion and advection of exudates and dye through the system. These equations also distinguish exudates in solution from dry exudates attached to soil particles. The model of the effect of root exudates on water transport is similar to that of Mair et al. (2025), but without incorporation of a root system that is actively taking up water and producing exudates.

Soil water transport is modelled using Richards equation (Richards 1931)

$$\partial_t\theta(h, c_W, c_D) - \nabla \cdot (K(h, c_W, c_D)\partial_x(h + x)) = 0 \quad x \in [-3.3, \quad 0], \quad t \in (0, T], \tag{2}$$

with the 1D spatial domain representing the depth of the chamber of transparent soil and the final time set to $T = 5$ days (120 h). As such, Equation (2) is solved for pressure head h cm and assumes that the movement of water content θ (cm^3 cm^{-3}) is driven by pressure gradients and gravity and is scaled according to the soil's hydraulic conductivity K (cm d^{-1}). To match the experimental conditions, the initial value of h was adjusted to -0.1 cm at the top layer of the chamber, -1000 cm in the middle and -0.1 cm on the bottom, which in terms of pressure approximately translates to -0.001 kPa, -100 kPa, and -0.001 kPa at the top middle and bottom layers respectively. The functions for water content and hydraulic conductivity were expressed from the soil pressure head using the formulations of (van Genuchten 1980) and (Mualem 1976):

$$\theta(h, c_W, c_D) = \theta_r + (\theta_s - \theta_r)S_e(h, c_W, c_D), \quad S_e(h, c_W, c_D) = \frac{1}{(1 + |\alpha(c_W, c_D)h|^n)^m}, \tag{3}$$

$$K(h, c_W, c_D) = K^* + \alpha_K K_s(c_W, c_D)K_r(h, c_W, c_D). \tag{4}$$

Both the Nafion particles and root exudates are known to have different behaviour during wetting and drying (hysteresis in both the hydraulic conductivity and water retention functions), and this was represented using the model of Kool and Parker (1987) for water content θ and the model of Vogel and Zhang (1996) for the hydraulic conductivity K. Hence, we focused primarily on the influence of the concentration of root exudates suspended in the solution c_W (mg mL^{-1}) and the concentration of root exudates dried and attached to the surface of the particles c_D (mg g^{-1}). These parameters affect the inverse air entry pressure head α (cm^{-1}), as described in (Karagunduz et al. 2001), through the surface tension of water γ (mN m^{-1})and water contact angle ω

$$\alpha^W = \alpha_0^W \times \frac{\gamma(c_W)}{\gamma_0} \times \frac{\cos(\omega_0)}{\cos\ (\omega(c_D))}, \quad \alpha^D = \alpha_0^D \times \frac{\gamma_0}{\gamma(c_W)}, \tag{5}$$

Here α^W and α^D are the inverse air entry pressure head for wetting and drying respectively and α_0^W and α_0^D are the default inverse air entry pressures if no exudates are present in either form. The effect of root exudates is also incorporated into the saturated hydraulic conductivity as

$$K_s^Z(c_W, c_D) = K_{s,0}^Z \times \left(\frac{\gamma_0}{\gamma(c_W)}\right)^\beta \times \frac{\cos(\omega(c_D))}{\cos(\omega_0)}, \tag{6}$$

where $Z = W, D$ for wetting or drying. Here $K_{s,0}^Z$ are the default saturated hydraulic conductivities during wetting and drying if no exudates are present, and β determines the extent to which an exudate-induced reduction in surface tension increases hydraulic conductivity. The terms K^* and α_K in (4) are included to effectively incorporate hysteresis. The expression for contact angle as functions of the concentrations of dried root exudates was fitted to the data of Zickenrott et al. (2016). Regarding the effect of root exudates on solution surface tension, we used the data set from Read et al. (2003) which provides solution surface tension values across a broad range of exudate concentrations. Moreover, at 1 mg mL^{-1} of exudates in solution the surface tension reported is similar to that which was obtained for the solution of tracer dye and root exudates in our own experiments. The following relationships for surface tension and contact angle were therefore obtained,

$$\gamma(c_W) = 47.5 + \frac{72.86 - 47.5}{(1 + e^{18.79(c_W - 0.36)})},$$

$$\omega(c_D) = \frac{3.529}{18}\pi\left(1 + e^{-1879.14c_D}\right)^{-4.39}. \tag{7}$$

The dynamics of the root exudates in solution and the dried root exudates are given by the following differential equations

$$\partial_t(\theta c_W) - \partial_x(\theta D_W \partial_x c_W - \vec{\boldsymbol{q}}\, c_W) = \rho\kappa_W c_D - \theta\kappa_D c_W,$$

$$\rho\partial_t c_D = \theta\kappa_D c_W - \rho\kappa_W c_D, \tag{8}$$

where D_W (cm^2 d^{-1}) is the coefficient of diffusion of root exudates through soil water, $\vec{\boldsymbol{q}}$ (cm d^{-1}) is the advective flux of soil water and ρ (mg cm^{-3}) is the bulk density of Nafion. The terms κ_W and κ_D (d^{-1}) are the rates at which dried root exudate joins the soil water solution and root exudate in solution dries to the pore surface respectively. Finally, the movement of dye within the chambers is modelled using an additional differential equation

$$\partial_t(\theta u) - \partial_x(\theta D_u \partial_x u - \vec{\boldsymbol{q}}\, u) = 0. \tag{9}$$

Here u (mL mL^{-1}) is the concentration of dye in solution and D_u (cm^2 d^{-1}) is the diffusion coefficient of the dye in solution. The dye concentration at the start of the experiment is 0.04 mL mL^{-1} in the upper layer where root exudates have been introduced and 0 in the middle and bottom layers.

A finite element scheme, with an implicit Euler discretisation in time, was used to approximate solutions to the coupled system of equations.

2.10 | Model Parameterisation

Model parameters were taken from previously published literature assuming that coarse sand would be the best approximation of the transparent soil. However, the reference inverse air-entry pressure heads α_0^D, α_0^W and saturated hydraulic conductivities $K_{s,0}^D, K_{s,0}^W$ were altered considerably from the values cited in the literature to reflect the hydrophobicity of dry Nafion that was observed in the control infiltration experiments. The full list of parameters can be found in Table 1.

Two types of critical parameters could not be determined from the literature. First, the coefficient linking surface tension to hydraulic conductivity β is very specific to the developed model system and hence could not be found in the literature. To calibrate the value of β, data was used from the second experiment and compared observations to simulated dye concentration between depths −3.3 cm and −1.8 cm at times $t = 3{,}4{,}5$ d. Using Bayesian optimisation (Brochu et al. 2010) we obtained the values β that most accurately matched experimental results. The rates at which root exudates dissolve or reverse to a dry state are also very specific to the model we used and were obtained in a similar way. Here we used the observation from the third experiment and Bayesian optimisation to determine the values of $\varkappa_W$ and $\varkappa_D$ that best predicted dye concentrations in the lower third of the domain. The model is available from the following Zenodo repository https://zenodo.org/records/16408615.

TABLE 1 | Model parameter values dervied from the scientific litterature.

Parameter	*Value*	*Source*
$\theta_{r,0}$	1×10^{-10} cm^3 cm^{-3}	$N\backslash A$
$\theta_{s,0}$	0.404 cm^3 cm^{-3}	Benson (2014)
α_0^D	0.049 cm^{-1}	Benson (2014)
α_0^W	0.164 cm^{-1}	Benson (2014)
n	3.16	Benson (2014)
m	0.684	Benson (2014)
$K_{s,0}^D$	8.196×10^{-4} cm d^{-1}	Carsel and Parrish (1988)
$K_{s,0}^W$	4.917×10^{-4} cm d^{-1}	Carsel and Parrish (1988)
D_W	0.65 cm^2 d^{-1}	Scott et al. 1995), average taken
ρ	1.9 g cm^{-3}	Oberbroeckling et al. (2002)
D_u	0.5 cm^2 d^{-1}	Syms (2017)

3 | Results

3.1 | Plant Roots Increase the Infiltration of Water Through Dry Transparent Soil in a Microcosm System

Root growth through the dry soil layer was substantially reduced due to the near complete absence of water. However, by the time the tracer dye was introduced, most samples had at least one root that had reached the base of the microcosm chamber. Following introduction into the microcosm, the dye was observed spreading non homogeneously but rapidly within the top transparent soil layer (Figure 2A), but there was no evidence of the dye being absorbed by the particles. The dry transparent soil layer blocked further spreading of the tracer dye into the lower layers of transparent soil. There was no obvious effect of the roots on the transport of the dye at this stage of the experiment. The spatial distribution of the tracer dye was analysed over a 72-h period following the start of the experiment (Figure 2B). In the control samples, the wetting front advanced downward by 1–2 mm during the first 48 h. We did not observe further progress of the wetting front during the rest of the experiment, even in the case where the tracer dye was seen in the bottom transparent soil layer (Figure 2B). 24 h after the introduction of the tracer dye, no coloration was observed in the bottom transparent soil layer of the control samples. Only five samples out of 13 showed some coloration at the end of the experiment. In the samples containing roots, we observed a greater movement of the wetting front. 48 h after the experiment began, the wetting front had advanced downward by 3 to 4 mm. When the root was visible, we observed the dye both attached to the root and in the surrounding transparent soil (Figure 2B). The tracer dye began to appear in the lower transparent soil layer 24 h after the introduction of the tracer dye. The coloration increased in intensity over time until the end of the experiments. All samples showed some coloration at the end of the experiment.

The quantitative analysis of the image data confirmed visual observations. The dye concentration in the dry transparent soil layer increased with time (Figure 2C). In the case of the control samples, the dye concentration increased up to a mean value of 3.5×10^{-4} mL mL^{-1}. In the case of the samples containing plants, an increase in dye concentration was observed during the entire experiment with a final dye concentration recorded at 2.1×10^{-2} mL mL^{-1}. Quantitatively, the estimated dye concentration in the dry transparent soil layer increased when a root was present, but not in control treatments (Figure 2C). We did not measure a significant change over time in the permeability of the dry transparent soil layer in the control samples (Figure 2D). Statistical analysis of the data using a linear model showed that in the absence of the root, the permeability of the dry transparent soil layer was not significantly different from 0 during the entire experiment ($p = 0.23$ and $p = 0.69$ for slope and intercept). However, the presence of the root had a significant effect on the rate of increase in the permeability of the dry transparent soil layer ($p < 0.001$).

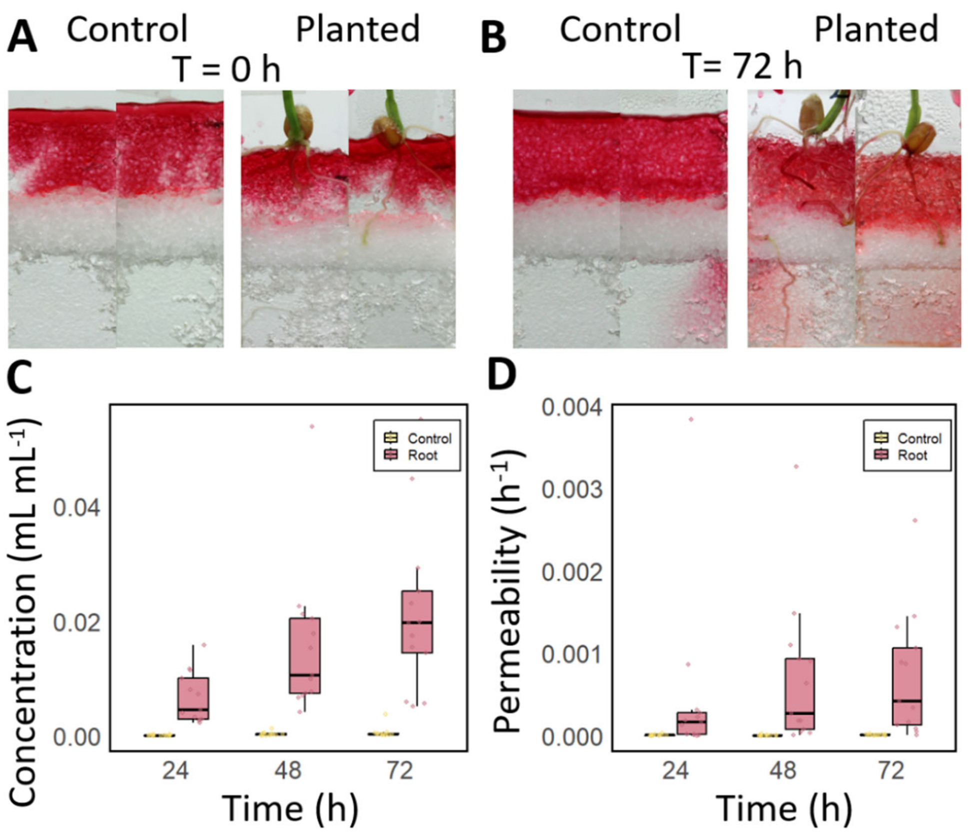

FIGURE 2 | Effect of the presence of winter wheat (*Triticum aestum*) roots on the infiltration of a tracer dye through a dry transparent soil layer. (A) Microcosm samples just after introduction of the tracer dye into the top transparent soil layer. Samples that did not contain roots (control) are shown on the left, and planted sample are shown on the right. (B) Microcosm samples 72 h after introduction of the tracer dye. (C) Throughout the experiment, the tracer dye concentration was higher in planted samples (red) than in control samples (yellow). (D) Throughout the experiment, the permeability of the dry soil layer was higher in planted samples (red) than in control samples (yellow). [Color figure can be viewed at wileyonlinelibrary.com]

3.2 | Root Exudates Increase Water Infiltration Through the Dry Transparent Soil Layer

Following introduction into the microcosm, the dye was observed spreading non homogeneously but rapidly within the top transparent soil layer (Figure 3A), but there was no evidence of the dye being absorbed by the transparent soil particles. The dry transparent soil layer blocked further spreading of the tracer dye into the lower layers of transparent soil. There was no obvious effect of the presence of root exudates on the transport of the dye at this stage of the experiment. The spatial distribution of the tracer dye was analysed over a 120-h period following the start of the experiment (Figure 3A,B). In control samples, the wetting front did not advance during the first 72 h but had advanced by 1–2 mm by the end of the experiment (Figure 3A,B). 96 h after the introduction of the tracer dye, red coloration became visible in the bottom transparent soil layer in all samples. The coloration increased in intensity with time until the end of the experiments. In the samples with root exudates, the wetting front advanced irregularly but continuously into the dry layer throughout the experiment. At the end of the experiment, the dry transparent soil layer was completely wet (Figure 3A,B).

Quantitatively, the mean dye concentration in the dry transparent soil layer increased with time (Figure 3C) for both control samples and samples containing root exudates. When samples contained root exudates, the increase in dye concentration in the bottom layer was sustained during the experiment. We did not measure a significant change with time in the permeability of the dry transparent soil layer in the control samples (Figure 3D). For samples with root exudates, permeability was maximal 120 h after introduction of the tracer dye. Statistical analysis of the data using a linear model showed that in the absence of root exudates, the permeability of the dry transparent soil layer was not significantly different from 0 during the entire experiment ($p = 0.54$ and $p = 0.45$ for slope and intercept). However, the presence of root exudates had a significant effect on the rate of increase in the permeability of the dry transparent soil layer ($p < 0.001$).

Measurement of surface tension showed that the presence of root exudates reduced the surface tension of solutions. Distilled water showed a value of 72.32 ± 0.11 mN m^{-1}, while the addition of exudates reduced the surface tension only slightly to 69.92 ± 0.47 mN m^{-1} at the concentration of 1 mg/mL and to 65.87 ± 0.80 mN/m at the concentration of 10 mg mL^{-1}. The tracing dye also influenced surface tension (68.24 ± 0.35 mN m^{-1}), but most interestingly, interacted with the presence of exudates in solution. The surface tension of solutions containing both surfactants and tracing dye was 54.19 ± 0.60 mN m^{-1}. Values are expressed as mean ± standard error. The surface

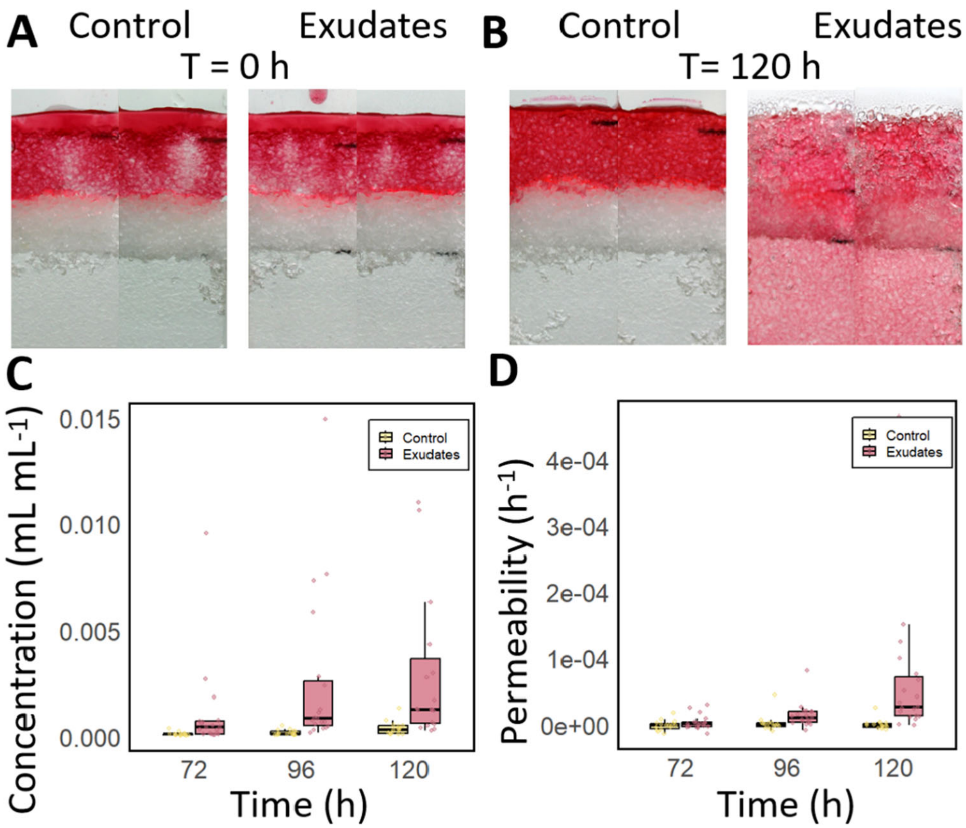

FIGURE 3 | Effect of the presence or absence of root exudates from winter wheat (*Triticum aestum*) on the infiltration of a tracer dye through a dry transparent soil layer. (A) Microcosm samples just after introduction of the tracer dye into the top transparent soil layer. Sample without root exudates (left) and sample with root exudates (right). (B) Microcosm samples 120 h after introduction the tracer dye. (C) Throughout the experiment, the tracer dye concentration was higher in samples containing exudates (red) than in control samples (yellow). (D) Throughout the experiment, the permeability of the dry soil layer was higher in samples containing exudates (red) than in control samples (green). [Color figure can be viewed at wileyonlinelibrary.com]

tension of wheat exudate solutions in the study of (Read et al. 2003) was approximately 50 mN m^{-1} at a concentration of 1 mg mL^{-1}, and therefore Equation 7 was calibrated on that data set.

3.3 | The Presence of Root Exudates in the Dry Layers Has Limited Effect on Water Infiltration and Rewetting of the Dry Transparent Soil Layer

Following introduction into the microcosm, the dye spread following a similar pattern as reported in the previous experiments (Figure 4A). There was no obvious effect of the presence of root exudates (in the top or dry layer) on the transport of the dye at this stage of the experiment. The spatial distribution of the tracer dye was analysed over a 120-h period following the start of the experiment (Figure 4A,B). In samples, that did not contain root exudates, the wetting front did not advance during the first 72 h but had advanced by 1–1.5 mm both in the NN treatment and in the NE treatments (Figure 3A,B). 96 h after the introduction of the tracer dye, red coloration became visible in the bottom transparent soil layer in all samples. The coloration increased in intensity with time until the end of the experiments. No colouration was observed at the bottom of the microcosm chambers at any stage of the experiment. There were no strong visual differences between samples with and without root exudates in the dry transparent soil layer. Samples containing root exudates in the tracer dye (EN and EE) exhibited qualitatively similar behaviour to samples containing root exudates in the tracer dye in experiment 2.

Quantitative analysis of the image data confirmed the visual observations. The dye concentration in the dry transparent soil layer increased with time (Figure 4C). For samples containing no exudates in the tracer dye, the increase was gradual, but for samples containing root exudates in the tracer dye, a sharp increase in dye concentration was observed within the first 72 h and this was followed by a more moderate increase in dye concentration. We did not measure a significant change over time in the permeability of the dry transparent soil layer in samples containing water in the tracer dye (Figure 4D). In the case of samples with root exudates in the top layer, permeability steadily increased during the experiment. Statistical analysis of the data using a linear model showed that in the absence of root exudates, the permeability of the dry transparent soil layer was not significantly different from 0 during the entire experiment but the presence of root exudates in the top transparent soil layer had both a significant effect on the slope and intercept of the relationship ($p < 0.001$ and $p = 0.01$ respectively). The presence of root exudates in the middle layer did not have a significant effect on the permeability of the dry transparent soil layer.

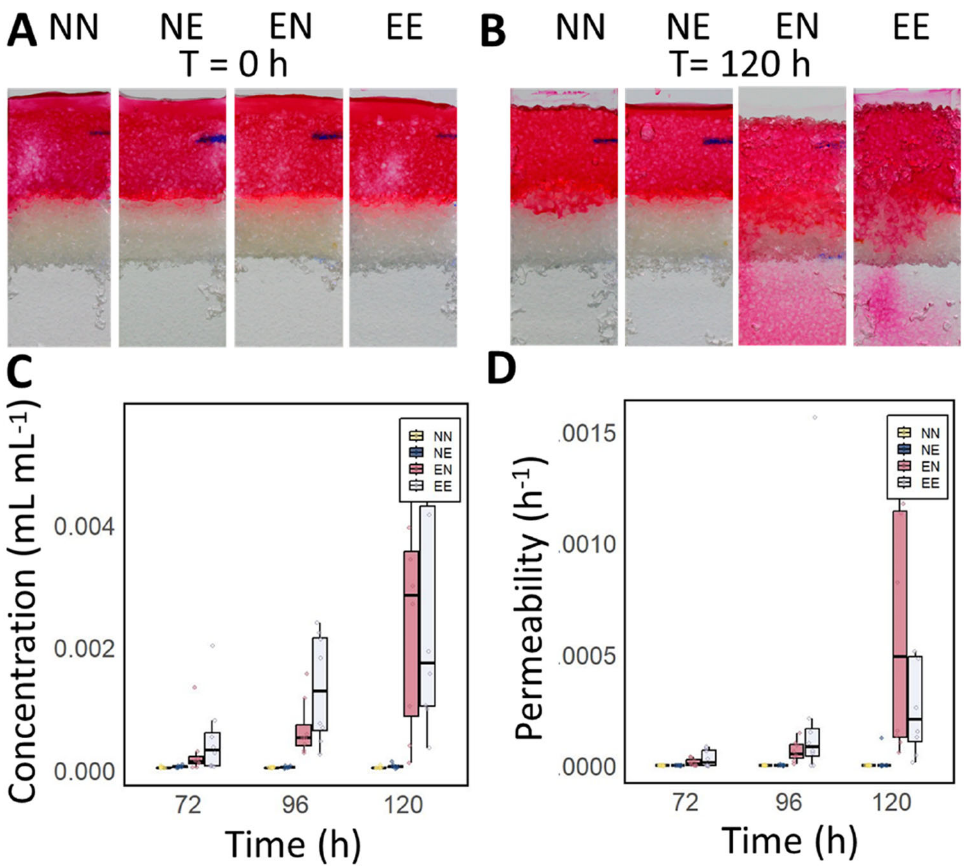

FIGURE 4 | Effect of location of root exudates of winter wheat (*Triticum aestum*) on the infiltration of a tracer dye through a dry transparent soil layer. (A) Microcosm samples just after introduction of the tracer dye into the top transparent soil layer. (B) Microcosm samples 120 h after the start of the experiment. (C) Concentration of the tracer dye in the dry transparent soil layer measured at different time points (from left to right: light yellow NN, dark green NE, dark red EN and light blue EE). (D) Permeability of the dry transparent soil layer measured at different time points (from left to right: light yellow NN, dark green NE, dark red EN and light blue EE). [Color figure can be viewed at wileyonlinelibrary.com]

3.4 | Physical Modifications Due to Penetration of the Dry Transparent Soil Layer Do Not Affect the Infiltration and Rewetting

Following introduction into the microcosm, the dye spread following a similar pattern as in the experiments where the tracer dye did not contain root exudates (Figure 5A,B). There was no obvious effect of the presence of the needle on the transport of the dye at this stage of the experiment. The spatial distribution of the tracer dye was analysed over a 120-h period following the start of the experiment. No difference was observed between treatments during the entire experiment. No dye was observed at the bottom transparent soil layer, nor in the dry transparent soil layer. At the end of the experiment, we observed that the wetting front had advanced about 1 mm downward for all treatments.

Quantitative analysis of the image data confirmed the visual observations. The dye concentration in the dry transparent soil layer did not increase with time under either of the conditions (Figure 5C). Calculation of the permeabilities of the dry transparent soil layer revealed that that when the needle was present, there was much higher variability and mean value for the permeability (Figure 5D), which is likely due to the needle having an effect on the saturation of the image. Statistical analyses however showed that the presence of a needle did not have an effect on the calculated permeability ($p = 0.0073$).

3.5 | The Hydraulic Conductivity Is Critically Sensitive to Surface Tension and Root Exudate Concentration

Model calibration was completed successfully with optimal parameters values for β^*, κ_D^*, κ_W^* determined from experimental data. Following calibration, the predicted dye concentration was plotted against experimental data (Figure 6). Results showed good agreements between the model and experimental observations. Discrepancies were observed in the control treatment where no exudates were introduced (NN). Experiments showed an abrupt stop in the permeability of the dry transparent soil layer, which the model could not represent. The estimated model parameters gave also interesting insights into the effects of root exudates on water transport in transparent soil. The exponent of the power law relating surface tension to hydraulic conductivity was $\beta = 8.8$, indicating that the concentration of root exudate in solution had a strong effect on the hydraulic conductivity of transparent soil. The parameter for the solubilisation or sorption of root exudates $\kappa_D^* = 9.9 \times 10^{-2} \ d^{-1}$ and $\kappa_W^* = 1.8 \times 10^{-2} d^{-1}$ also provided useful insight. They showed that although the rate of solubilisation may not be limiting, because permeability is very small at the start of the experiments, the fraction of the transparent soil exposed to water limits the capacity of the dry root exudates to meaningfully increase the infiltration of water.

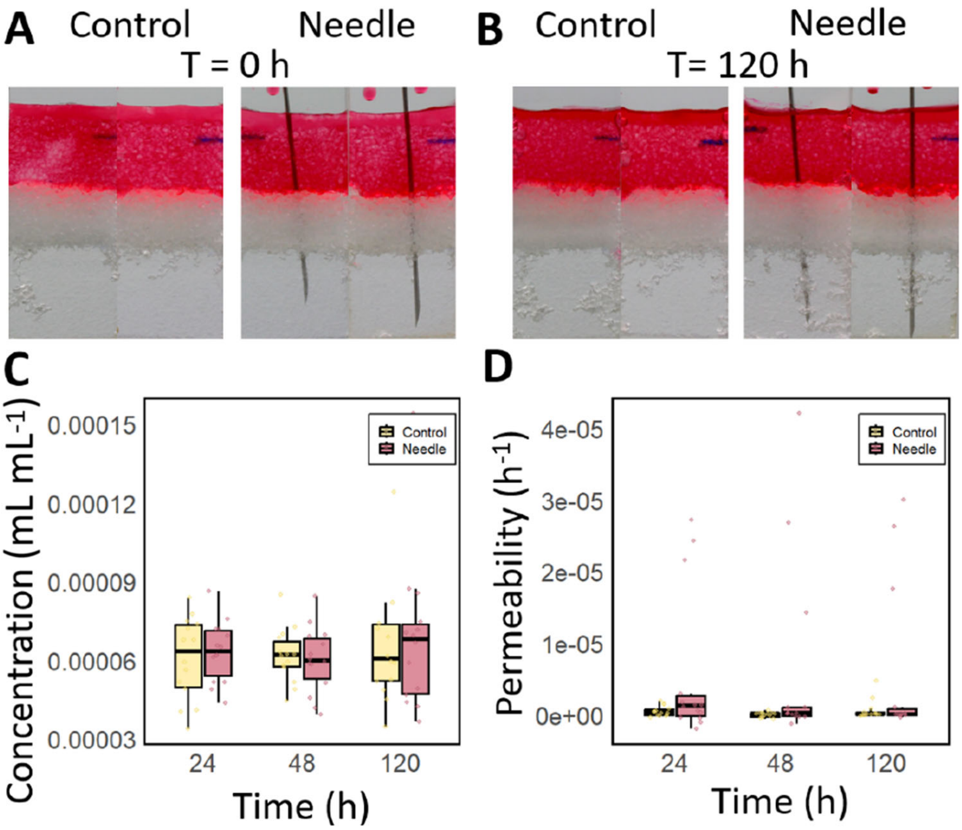

FIGURE 5 | Effect of the presence or absence of a needle on the infiltration of a tracer dye through a dry transparent soil layer. (A) Microcosm samples just after introduction of the tracer dye into the top transparent soil layer. Sample without needle (Control, left) and sample with needle (right). (B) Microcosm samples 120 h after introduction of the tracer dye. (C) Concentration of the tracer dye in the dry transparent soil layer measured at different time points (yellow control samples, red samples with a needle). (D) Permeability of the dry transparent soil layer measured at different time points (green control samples, pink samples with a needle). [Color figure can be viewed at wileyonlinelibrary.com]

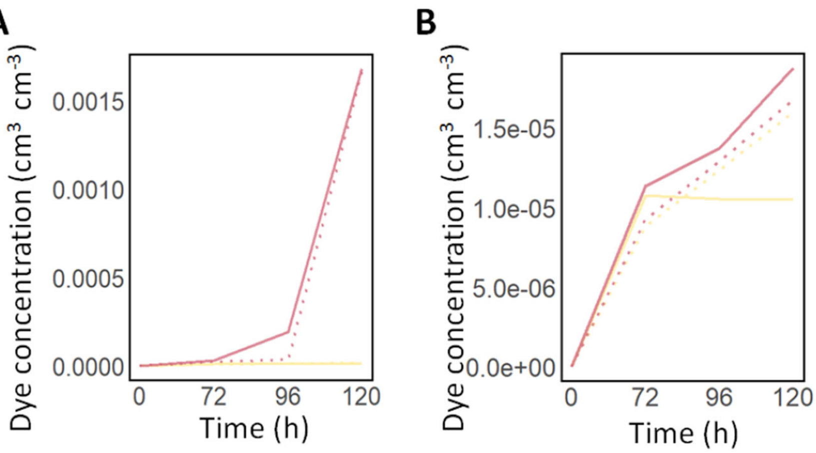

FIGURE 6 | Model predictions against experimental data for dye concentration in the bottom transparent soil layer. (A) Comparison between model and experimental data for experiments when root exudates are introduced in the top layer in the tracer dye solution. EN is shown in red and NN is shown in yellow. (B) Comparison between model and experimental data for experiments when root exudates are introduced dry in the middle layer. NE is shown in red and NN is shown in yellow. In all plots, dotted lines represent model predictions data whereas plain lines indicate experimental observations. [Color figure can be viewed at wileyonlinelibrary.com]

4 | Discussion

4.1 | Water Infiltration and Loss of Water From the Rhizosphere

The root system architecture, the spatial and topological arrangement of roots into a soil, is a critical trait to determine the ability of a crop to extract water from the soil. Firstly, because it determines the regions of the soil where water can be extracted. When a root system is shallow, the plant can more readily acquire immobile nutrients that are strongly absorbed to soil particles (Lynch and Brown 2001; Yi et al. 2022). When the root system is deep, it can reach the water table and extract groundwater during drought (Lynch 2013; Feng et al. 2022). However, root system architecture is highly dynamic, continuously adapting to developmental cues and environmental conditions – challenging the traditional concept of a fixed root ideotype (Tardieu et al. 2017). For example, root hydraulic conductivity has long been known to vary with age (Doussan 1998). Root hairs, which play a key role in the absorption of water and nutrients, are believed to have a short lifespan – typically ranging from a few days to about 3 weeks after emergence (Cai and Ahmed 2022). Numerous anatomical and biochemical properties of roots involved in resource uptake are also known to be plastic in response to both biotic and abiotic stresses (Lynch et al. 2021; Shukla and Barberon 2021).

Since the water balance in soil is very dynamic, another important feature of root systems may be their ability to intercept water, that is to distribute water into soil following a rainfall, to minimise losses from the rhizosphere. Although roots are known to facilitate preferential water flow through the soil (Jačka et al. 2021), this function of root systems has been largely overlooked as a valuable trait to improve water management in cropping systems. Recently, mathematical models that combined water uptake with preferential flow induced by plant roots showed that root system architectures may have very different abilities to intercept rainfall water, and that root system ideotypes may vary as a function of soil type (Mair et al. 2023).

The experiments described in this study aimed to investigate the impact of root traits on water infiltration in dry soil. We used a dye tracing approach rather than infiltrometers to miniaturise and shorten the duration of the microcosm experiments. Observation of the dye also gave insights into the mechanisms of infiltration. The use of an artificial soil improved the accuracy of measurements since all the dye below the dry layer was considered in the measurement of the fluxes. The artificial soils also made experiments more manageable. Nafion, the material used as substrate, is mildly hydrophobic when dry (water contact angle of 105° [Goswami et al. 2008]), and resulted in low infiltration rates but with measurable effects after a few days of experiments. Agricultural soils which have low organic matter content usually exhibit higher infiltration rates (Basset et al. 2023), and organic matter rich soils such as peat are too hydrophobic when dry (water contact angle of up to 120° [Valat et al. 1991]).

4.2 | Physical Processes Involved in Preferential Flow Induced by Plant Roots

We observed that root exudation was a primary factor influencing the infiltration of water through dry soil. This result suggests that surfactants found in root exudates may play a crucial role in affecting water movement within the soil. Not only are root exudates known to contain surfactants (Read et al. 2003), but measurements of water retention in rhizosphere and bulk soil have also indicated that rhizosphere solutions may exhibit lower surface tension (Whalley et al. 2005). Surfactants are also used for the irrigation of plants growing in hydrophobic soils and arid environments (Baratella and Trinchera 2018; Ogunmokun et al. 2020), which confirms that the exudation of surfactants may be a beneficial trait in certain soils.

Experiments using syringe needles indicated that changes in porosity induced by root growth may not significantly affect the infiltration of water. The closest natural soil conditions to our experimental system is a coarse sand. When loosely packed, root growth was shown to induce dilation of $< 4\%$ and increase the porosity of about 10% over a layer of < 2 mm around the root (Anselmucci et al. 2021). The Kozeny–Carman equation $K \propto \text{porosity}^3/(1-\text{porosity})^2$ predicts that such changes can result in an increase of 50% of the hydraulic conductivity (Schulz et al. 2019). But a 1 mm thick layer represents about 10% of the cross section of the soil. Also, dilation of the rhizosphere soil is associated with reduction of porosity further away from the root and collectively, this would likely limit the changes in the hydraulic conductivity to less than 5%. In the Young–Laplace equation, pore size and surface tension exert equal effects on capillary entry pressure (Hallett 2008). Given that root exudates have been shown to reduce surface tension by 27%, reducing it from 72 mN m^{-1} to as low a 45 mN m^{-1} (Read et al. 2003; Naveed et al. 2019), it is plausible for root exudation to have a stronger effect than dilation in some conditions.

Our measurements of the surface tension of root exudate solutions were higher than the values reported in previous studies. Surface tensions as low as 45 mN m^{-1} were reported on maize mucilage solution at 1 mg mL^{-1} when extraction was performed directly on the root tips with a pipette (Read and Gregory 1997). Naveed et al. (2019) reported that wheat root exudate solutions extracted from hydroponic culture have a surface tension of approximately 55 mN m^{-1} when at a concentration of 1 mg mL^{-1}. These differences could be explained by factors such as the age and the species studied, or the duration of the extraction. However, the extraction method we employed – which samples the hydroponic solution rather than collecting it in its entirety – may discard a substantial fraction of surfactants which accumulate at air–water interfaces. Although our experiments were not performed under strictly axenic conditions. Therefore, it is possible that bacterial secretions or motility (Saintillan 2018; Benard et al. 2023) influenced the results.

The magnitude of root exudates' facilitation of water infiltration when in natural soils, however, remain largely uncertain. The substrate used in this study, a granular made of Nafion polymer, has properties which cannot be commonly found in a natural environment since it has the texture of a coarse sand but when dry has a contact angle closer to that of peat (Valat et al. 1991). Using a mathematical model (Equations 2–9), we also found that to match experimental observations of water infiltration, not only must the pore entry pressure change in the presence of exudates (controlled by surface tension) but the hydraulic conductivity coefficient must increase too. These results indicate that the reduction in surface tension alone may not be sufficient to explain the facilitation of water transport. Jurin's law of capillary action, which relates surface tension to capillary entry pressure, describes an equilibrium (Hallett 2008), whereas infiltration is a slow but dynamic process. The shear rates through which water is moving are likely extremely slow and root exudates and mucilage can exhibit variations in viscosity depending on the shear rate of the flow, possibly enhancing infiltrations in certain conditions (Read et al. 1999; Carminati et al. 2017; Naveed et al. 2019).

4.3 | Influence of Root Exudation on Soil Moisture Redistribution

During drought conditions, evidence suggests that both the quantity and composition of root exudates undergo significant changes (Karlowsky et al. 2018; Williams and de Vries 2020). However, it remains uncertain whether these changes reflect a plant's response to modify the physical properties of the soil. It was suggested that for a large part, changes in root exudation result from shifts in a plant's metabolic activity, with root exudates reflecting the most abundant root metabolites (Gargallo-Garriga et al. 2018). The lifespan of root exudates in soils is likely to be highly variable. Many of the compounds deposited in the soil by plant roots are consumed by soil microbes (Sasse et al. 2018), and it is challenging to determine the extent to which changes in soil physical properties are directly attributable to rhizodeposition, as opposed to being secondary effects arising from microbial activity. For example, soil bacteria are producers of potent biosurfactants (Karanth et al. 1999; Domingues et al. 2020). They are also able to synthesise extracellular polymeric substances (EPS) which enhance soil water retention (Zheng et al. 2018; Benard et al. 2023). The reduction of surface tension typically necessitates small amounts of a single surfactant molecule, such as Lecithin found in soybeans (Read et al. 2003). Consequently, even subtle changes in the composition of root exudates could significantly impact soil properties.

For these reasons, it is unclear what the desired properties of the rhizosphere solution should be. It also remains unclear which physical properties the soil solution must exhibit in the rhizosphere to support optimal conditions. In a cropping system, evaporation is a main source of water loss with typically 15%–70% of the evapotranspiration lost due to evaporation (Hernández et al. 2015; Unkovich et al. 2018). In sandy soils however, deep percolation may become the dominant source of water loss and increased infiltration would impact the crop negatively. Nassah et al. (2018) reported nearly 50% loss of water inputs lost due to water leaving the rooted zone through deep percolation. Surfactant production may be desirable in low soil water content when water is trapped in small pores (Whalley et al. 2005) or in soils rich in organic matter (strong hysteresis between wetting and drying), but the broader benefits are less clear. The level of organic matter in agricultural soils is minimal (Loveland 2003; Rawls et al. 2004) and heterogeneously distributed (Lehmann et al. 2008). Also, losses due to evaporation and deep percolation are extremely dependent on rainfall or irrigation patterns.

Overall, differences in exudate composition and concentration, microbial activity, and the unique combination of pore connectivity and soil particle surface chemistry in a natural soil are likely to interact in complex and yet unknown ways to influence the infiltration of water. Understanding which root exudate properties enhance plant water availability requires not only the development of novel experimental systems capable of measuring the effects of root exudation on water infiltration in natural soils, but also the integration of mathematical modelling approaches to disentangle the many interacting factors influencing this process.

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Data Availability Statement

The data that support the findings of this study are openly available at https://doi.org/10.6084/m9.figshare.29194898.v1 https://doi.org/10.6084/m9.figshare.29136761.v1. The model is available from the following Zenodo repository https://zenodo.org/records/16408615.

References

Ahmed, M. A., E. Kroener, P. Benard, M. Zarebanadkouki, A. Kaestner, and A. Carminati. 2016. "Drying of Mucilage Causes Water Repellency in the Rhizosphere of Maize: Measurements and Modelling." *Plant and Soil* 407: 161–171.

Anselmucci, F., E. Andò, G. Viggiani, N. Lenoir, C. Arson, and L. Sibille. 2021. "Imaging Local Soil Kinematics During the First Days of Maize Root Growth in Sand." *Scientific Reports* 11, no. 1: 22262.

Baratella, V., and A. Trinchera. 2018. "Organosilicone Surfactants as Innovative Irrigation Adjuvants: Can They Improve Water Use Efficiency and Nutrient Uptake in Crop Production?" *Agricultural Water Management* 204: 149–161.

Basset, C., M. Abou Najm, T. Ghezzehei, X. Hao, and A. Daccache. 2023. "How Does Soil Structure Affect Water Infiltration? A Meta-Data Systematic Review." *Soil and Tillage Research* 226: 105577.

Benard, P., S. Bickel, A. Kaestner, P. Lehmann, and A. Carminati. 2023. "Extracellular Polymeric Substances From Soil-Grown Bacteria Delay Evaporative Drying." *Advances in Water Resources* 172: 104364.

Benson, C. H., et al. 2014. "Estimating Van Genuchten Parameters α and n for Clean Sands From Particle Size Distribution Data." In *From Soil Behavior Fundamentals to Innovations in Geotechnical Engineering: Honoring Roy E. Olson*, 410–427. American Society of Civil Engineers.

Berry, J. D., M. J. Neeson, R. R. Dagastine, D. Y. C. Chan, and R. F. Tabor. 2015. "Measurement of Surface and Interfacial Tension Using Pendant Drop Tensiometry." *Journal of Colloid and Interface Science* 454: 226–237.

Brochu, E., V. M. Cora, and N. De Freitas 2010. 'A Tutorial on Bayesian Optimization of Expensive Cost Functions, With Application to Active User Modeling and Hierarchical Reinforcement Learning', *arXiv preprint arXiv:1012.2599* [Preprint].

Bruand, A., I. Cousin, B. Nicoullaud, O. Duval, and J. C. Bégon. 1996. "Backscattered Electron Scanning Images of Soil Porosity for Analyzing Soil Compaction Around Roots." *Soil Science Society of America Journal* 60, no. 3: 895–901.

Cai, G., and M. A. Ahmed. 2022. "The Role of Root Hairs in Water Uptake: Recent Advances and Future Perspectives." *Journal of Experimental Botany* 73, no. 11: 3330–3338.

Carminati, A., P. Benard, M. A. Ahmed, and M. Zarebanadkouki. 2017. "Liquid Bridges at the Root-Soil Interface." *Plant and Soil* 417: 1–15.

Carminati, A., A. B. Moradi, D. Vetterlein, et al. 2010. "Dynamics of Soil Water Content In the Rhizosphere." *Plant and Soil* 332, no. 1–2: 163–176. https://doi.org/10.1007/s11104-010-0283-8.

Carsel, R. F., and R. S. Parrish. 1988. "Developing Joint Probability Distributions of Soil Water Retention Characteristics." *Water Resources Research* 24, no. 5: 755–769.

Clothier, B. E., and S. R. Green. 1997. "Roots: The Big Movers of Water and Chemical in Soil." *Soil Science* 162, no. 8: 534–543.

Dexter, A. R. 1987. "Compression of Soil Around Roots." *Plant and Soil* 97: 401–406.

Dobson, A. J., and A. G. Barnett. 2018. *An Introduction to Generalized Linear Models*. Chapman and Hall/CRC.

Domingues, V. S., A. de Souza Monteiro, A. D. L. Júlio, A. L. L. Queiroz, and V. L. dos Santos. 2020. "Diversity of Metal-Resistant and Tensoactive-Producing Culturable Heterotrophic Bacteria Isolated From a Copper Mine in Brazilian Amazonia." *Scientific Reports* 10, no. 1: 6171.

Doussan, C. 1998. "Modelling of the Hydraulic Architecture of Root Systems: An Integrated Approach to Water Absorption—Distribution of Axial and Radial Conductances In Maize." *Annals of Botany* 81, no. 2: 225–232.

Downie, H., N. Holden, W. Otten, A. J. Spiers, T. A. Valentine, and L. X. Dupuy. 2012. "Transparent Soil for Imaging the Rhizosphere." *PLoS One* 7, no. 9: e44276. https://doi.org/10.1371/journal.pone.0044276.

Duddek, P., A. Carminati, N. Koebernick, et al. 2022. "The Impact of Drought-Induced Root and Root Hair Shrinkage on Root–Soil Contact." *Plant Physiology* 189, no. 3: 1232–1236.

Feng, X., L. Jia, Y. Cai, et al. 2022. "Aba-Inducible Deeper Rooting 1 Improves Adaptation of Maize to Water Deficiency." *Plant Biotechnology Journal* 20, no. 11: 2077–2088.

Gargallo-Garriga, A., C. Preece, J. Sardans, M. Oravec, O. Urban, and J. Peñuelas. 2018. "Root Exudate Metabolomes Change Under Drought and Show Limited Capacity for Recovery." *Scientific Reports* 8, no. 1: 12696.

van Genuchten, M. T. 1980. "A Closed-Form Equation for Predicting the Hydraulic Conductivity of Unsaturated Soils." *Soil Science Society of America Journal* 44, no. 5: 892–898. https://doi.org/10.2136/sssaj1980.03615995004400050002x.

Goswami, S., S. Klaus, and J. Benziger. 2008. "Wetting and Absorption of Water Drops on Nafion Films." *Langmuir* 24, no. 16: 8627–8633.

Hallam, J., J. Holden, D. A. Robinson, and M. E. Hodson. 2021. "Effects of Winter Wheat and Endogeic Earthworms on Soil Physical and Hydraulic Properties." *Geoderma* 400: 115126. https://doi.org/10.1016/j.geoderma.2021.115126.

Hallett, P. D. 2008. "A Brief Overview of the Causes, Impacts and Amelioration of Soil Water Repellency–A Review." *Soil and Water Research* 3, no. 1: S21–S29.

Helliwell, J. R., C. J. Sturrock, S. Mairhofer, et al. 2017. "The Emergent Rhizosphere: Imaging the Development of the Porous Architecture at the Root-Soil Interface." *Scientific Reports* 7, no. 1: 14875.

Hernández, M., L. Echarte, A. Della Maggiora, M. Cambareri, P. Barbieri, and D. Cerrudo. 2015. "Maize Water Use Efficiency and Evapotranspiration Response to N Supply Under Contrasting Soil Water Availability." *Field Crops Research* 178: 8–15.

Jačka, L., A. Walmsley, M. Kovář, and J. Frouz. 2021. "Effects of Different Tree Species on Infiltration and Preferential Flow In Soils Developing at a Clayey Spoil Heap." *Geoderma* 403: 115372.

Karagunduz, A., K. D. Pennell, and M. H. Young. 2001. "Influence of a Nonionic Surfactant on the Water Retention Properties of Unsaturated Soils." *Soil Science Society of America Journal* 65, no. 5: 1392–1399. https://doi.org/10.2136/sssaj2001.6551392x.

Karanth, N. G. K., P. G. Deo, and N. K. Veenanadig. 1999. "Microbial Production of Biosurfactants and Their Importance." *Current Science* 77, no. 1: 116–126.

Karlowsky, S., A. Augusti, J. Ingrisch, M. K. U. Akanda, M. Bahn, and G. Gleixner. 2018. "Drought-Induced Accumulation of Root Exudates Supports Post-Drought Recovery of Microbes In Mountain Grassland." *Frontiers in Plant Science* 9: 1593.

Koebernick, N., K. R. Daly, S. D. Keyes, et al. 2019. "Imaging Microstructure of the Barley Rhizosphere: Particle Packing and Root Hair Influences." *New Phytologist* 221, no. 4: 1878–1889.

Kool, J. B., and J. C. Parker. 1987. "Development and Evaluation of Closed-Form Expressions for Hysteretic Soil Hydraulic Properties." *Water Resources Research* 23, no. 1: 105–114. https://doi.org/10.1029/WR023i001p00105.

Lehmann, J., D. Solomon, J. Kinyangi, L. Dathe, S. Wirick, and C. Jacobsen. 2008. "Spatial Complexity of Soil Organic Matter Forms at Nanometre Scales." *Nature Geoscience* 1, no. 4: 238–242.

Leung, A. K., et al. 2018. "Plant Age Effects on Soil Infiltration Rate During Early Plant Establishment." *', Géotechnique* 68, no. 7: 646–652.

Liu, Y., D. Patko, A. Lora de la Mata, et al. 2025. "Microcosm Fabrication Platform for Live Microscopy of Plant-Soil Systems." *Biosystems Engineering* 252: 105–114.

Loveland, P. 2003. "Is There a Critical Level of Organic Matter in the Agricultural Soils of Temperate Regions: A Review." *Soil and Tillage research* 70, no. 1: 1–18.

Lucas, M., S. Schlüter, H. J. Vogel, and D. Vetterlein. 2019. "Roots Compact the Surrounding Soil Depending on the Structures They Encounter." *Scientific Reports* 9, no. 1: 16236.

Lynch, J. P. 2013. "Steep, Cheap and Deep: An Ideotype to Optimize Water and N Acquisition by Maize Root Systems." *Annals of Botany* 112, no. 2: 347–357.

Lynch, J. P., and K. M. Brown. 2001. "Topsoil Foraging - an Architectural Adaptation of Plants to Low Phosphorus Availability." *Plant and Soil* 237: 225–237.

Lynch, J. P., C. F. Strock, H. M. Schneider, et al. 2021. "Root Anatomy and Soil Resource Capture." *Plant and Soil* 466: 21–63.

Mair, A., E. G. Peral, M. Ptashnyk, and L. Dupuy. 2025. "Modelling the Influence of Rhizodeposits on Root Water Uptake." *bioRxiv*. https://doi.org/10.1101/2025.03.28.645940.

Mair, A., L. Dupuy, and M. Ptashnyk. 2023. "Can Root Systems Redistribute Soil Water to Mitigate the Effects of Drought?" *Field Crops Research* 300: 109006. https://doi.org/10.1016/j.fcr.2023.109006.

Marshall, M. R., C. E. Ballard, Z. L. Frogbrook, et al. 2014. "The Impact of Rural Land Management Changes on Soil Hydraulic Properties and Runoff Processes: Results From Experimental Plots In Upland UK." *Hydrological Processes* 28, no. 4: 2617–2629.

Mualem, Y. 1976. "A New Model for Predicting the Hydraulic Conductivity of Unsaturated Porous Media." *Water Resources Research* 12, no. 3: 513–522. https://doi.org/10.1029/WR012i003p00513.

Nassah, H., S. Er-Raki, S. Khabba, et al. 2018. "Evaluation and Analysis of Deep Percolation Losses of Drip Irrigated Citrus Crops Under Non-Saline and Saline Conditions In a Semi-Arid Area." *Biosystems Engineering* 165: 10–24.

Naveed, M., M. A. Ahmed, P. Benard, et al. 2019. "Surface Tension, Rheology and Hydrophobicity of Rhizodeposits and Seed Mucilage Influence Soil Water Retention and Hysteresis." *Plant and Soil* 437: 65–81.

Ni, J. J., A. K. Leung, and C. W. W. Ng. 2019. "Modelling Effects of Root Growth and Decay on Soil Water Retention and Permeability." *Canadian Geotechnical Journal* 56, no. 7: 1049–1055.

Oberbroeckling, K. J., D. C. Dunwoody, S. D. Minteer, and J. Leddy. 2002. "Density of Nafion Exchanged With Transition Metal Complexes and Tetramethyl Ammonium, Ferrous, and Hydrogen Ions: Commercial and Recast Films." *Analytical Chemistry* 74, no. 18: 4794–4799.

Oburger, E., and D. L. Jones. 2018. "Sampling Root Exudates–Mission Impossible?" *Rhizosphere* 6: 116–133.

Ogunmokun, F. A., Z. Liu, and R. Wallach. 2020. "The Influence of Surfactant-Application Method on the Effectiveness of Water-Repellent Soil Remediation." *Geoderma* 362: 114081.

R Core Team. 2021. *R Core Team (2021). R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing.

Rawls, W. J., A. Nemes, and Y. A. Pachepsky. 2004. "Effect of Soil Organic Carbon on Soil Hydraulic Properties." *Developments in Soil Science* 30: 95–114.

Read, D. B., A. G. Bengough, P. J. Gregory, et al. 2003. "Plant Roots Release Phospholipid Surfactants That Modify the Physical and Chemical Properties of Soil." *New Phytologist* 157, no. 2: 315–326.

Read, D. B., and P. J. Gregory. 1997. "Surface Tension and Viscosity of Axenic Maize and Lupin Root Mucilages." *New Phytologist* 137, no. 4: 623–628.

Read, D. B., P. J. Gregory, and A. E. Bell. 1999. "Physical Properties of Axenic Maize Root Mucilage." *Plant and Soil* 211: 87–91.

Richards, L. A. 1931. "Capillary Conduction of Liquids Through Porous Mediums." *Physics* 1, no. 5: 318–333.

Saintillan, D. 2018. "Rheology of Active Fluids." *Annual Review of Fluid Mechanics* 50, no. 1: 563–592.

Sasse, J., E. Martinoia, and T. Northen. 2018. "Feed Your Friends: Do Plant Exudates Shape the Root Microbiome?." *Trends in Plant Science* 23, no. 1: 25–41.

Schmid, B., J. Schindelin, A. Cardona, M. Longair, and M. Heisenberg. 2010. "A High-Level 3D Visualization Api for Java and ImageJ." *BMC Bioinformatics* 11, no. 1: 274. http://www.biomedcentral.com/1471-2105/11/274.

Schulz, R., N. Ray, S. Zech, A. Rupp, and P. Knabner. 2019. "Beyond Kozeny–Carman: Predicting the Permeability in Porous Media." *Transport in Porous Media* 130: 487–512.

Scott, E. M., E. A. S. Rattray, J. I. Prosser, et al. 1995. "A Mathematical Model for Dispersal of Bacterial Inoculants Colonizing the Wheat Rhizosphere." *Soil Biology and Biochemistry* 27, no. 10: 1307–1318.

Shukla, V., and M. Barberon. 2021. "Building and Breaking of a Barrier: Suberin Plasticity and Function In the Endodermis." *Current Opinion in Plant Biology* 64: 102153.

Song, L., J. H. Li, T. Zhou, and D. G. Fredlund. 2017. "Experimental Study on Unsaturated Hydraulic Properties of Vegetated Soil." *Ecological Engineering* 103: 207–216.

Syms, R. 2017. "Rapid Evaporation-Driven Chemical Pre-Concentration and Separation on Paper." *Biomicrofluidics* 11, no. 4: 044116.

Tardieu, F., X. Draye, and M. Javaux. 2017. "Root Water Uptake and Ideotypes of the Root System: Whole-Plant Controls Matter." *Vadose Zone Journal* 16, no. 9: 1–10.

Unkovich, M., J. Baldock, and R. Farquharson. 2018. "Field Measurements of Bare Soil Evaporation and Crop Transpiration, and Transpiration Efficiency, for Rainfed Grain Crops In Australia–A Review." *Agricultural Water Management* 205: 72–80.

Valat, B., C. Jouany, and L. M. Riviere. 1991. "Characterization of the Wetting Properties of Air-Dried Peats and Composts." *Soil Science* 152, no. 2: 100–107.

Vogel, T., and R. Zhang 1996. 'The HYDRUS Code for Simulating One-Dimensional Water Flow, Solute Transport, and Heat Movement in Variably-Saturated Media. Version 5.0'. https://doi.org/10.13140/RG.2.1.3456.7525.

Whalley, W. R., B. Riseley, P. B. Leeds-Harrison, N. R. A. Bird, P. K. Leech, and W. P. Adderley. 2005. "Structural Differences Between Bulk and Rhizosphere Soil." *European Journal of Soil Science* 56, no. 3: 353–360.

Williams, A., and F. T. de Vries. 2020. "Plant Root Exudation Under Drought: Implications for Ecosystem Functioning." *New Phytologist* 225, no. 5: 1899–1905.

Xiao, T., P. Li, W. Fei, and J. Wang. 2024. "Effects of Vegetation Roots on the Structure and Hydraulic Properties of Soils: A Perspective Review." *Science of the Total Environment* 906: 167524.

Yi, K., X. Li, D. Chen, et al. 2022. "Shallower Root Spatial Distribution Induced by Phosphorus Deficiency Contributes to Topsoil Foraging and Low Phosphorus Adaption in Sugarcane (*Saccharum officinarum* L.)." *Frontiers in Plant Science* 12: 797635.

Zheng, W., S. Zeng, H. Bais, et al. 2018. "Plant Growth-Promoting Rhizobacteria (PGPR) Reduce Evaporation and Increase Soil Water Retention." *Water Resources Research* 54, no. 5: 3673–3687.

Zhou, H., W. R. Whalley, M. J. Hawkesford, et al. 2021. "The Interaction Between Wheat Roots and Soil Pores in Structured Field Soil." *Journal of Experimental Botany* 72, no. 2: 747–756. https://doi.org/10.1093/jxb/eraa475.

Zickenrott, I. M., S. K. Woche, J. Bachmann, M. A. Ahmed, and D. Vetterlein. 2016. "An Efficient Method for the Collection of Root Mucilage From Different Plant Species—A Case Study on the Effect of Mucilage on Soil Water Repellency." *Journal of Plant Nutrition and Soil Science* 179, no. 2: 294–302. https://doi.org/10.1002/jpln.201500511.

Supporting Information

Additional supporting information can be found online in the Supporting Information section.
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